

AYULEAF VISION:

AI-Based System for Ayurvedic Plant Disease Detection & Analysis

Intelligent Disease Diagnosis Pipeline using MobileNetV2, CLAHE, HSV Segmentation & Grad-CAM

PRESENTED BY

Akaash K (22BCE5218) | S R Kousalya (22BCE5225)

Guide:

Dr. Suseela S

Neem

Tulsi

Turmeric

Aloe Vera

PROJECT OVERVIEW

What We Built

An end-to-end AI-powered diagnostic pipeline for Ayurvedic plant disease detection — combining deep learning classification, lesion segmentation, explainability, and environmental analysis into a single web-deployable system.

Why It Matters

Farmers & herbalists lack access to expert diagnostics. Traditional visual inspection is error-prone, unscalable, and unavailable in rural areas. Our solution brings lab-grade AI diagnosis to any smartphone.

Target Ayurvedic Plants

- Neem:** Antibacterial, antifungal — 6 disease classes
- Tulsi:** Respiratory & immune — 3 disease classes
- Turmeric:** Anti-inflammatory (curcumin) — 2 disease classes
- Aloe Vera:** Skincare & digestive — 4 disease classes

13,460

Leaf Images

18

Disease Classes

4

Plant Species

90.13%

Test Accuracy

89.87%

Weighted F1 Score

9.25 MB

Model Size

PROBLEM STATEMENT & OBJECTIVES

No Ayurvedic-Specific Datasets

Existing models target food crops. Neem, Tulsi, Turmeric & Aloe Vera are underserved.

Black-Box Classification

Current systems produce labels with no spatial explanation — farmers can't trust what they can't see.

No Severity Quantification

Knowing the disease name is insufficient. Treatment urgency depends on lesion extent.

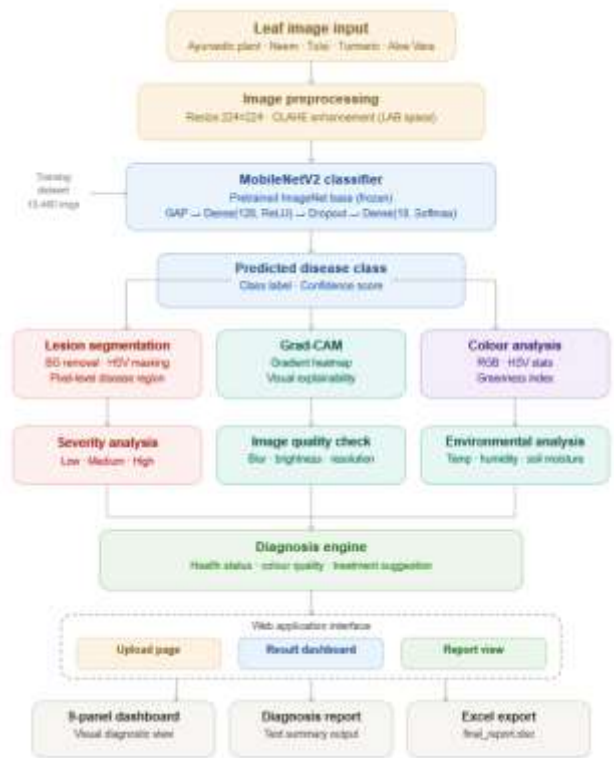
Ignores Environmental Context

Field conditions (temp, humidity, soil) directly affect disease risk — ignored by existing tools.

PROJECT OBJECTIVES

- 1 Train MobileNetV2 classifier on 18 disease classes with >85% accuracy
- 2 Implement CLAHE enhancement in LAB colour space for better lesion visibility
- 3 Build HSV-based lesion segmentation for pixel-level severity (Low / Medium / High)
- 4 Integrate Grad-CAM to generate spatial explainability heatmaps
- 5 Add environmental analysis module (temperature, humidity, soil moisture)
- 6 Export structured Excel diagnosis reports with treatment recommendations

SYSTEM DESIGN AND ARCHITECTURE



End-to-End Intelligent Diagnostic Pipeline



Lesion Segmentation

- Convert to HSV colour space
- Threshold brownish-yellow pixels [10–35 hue]
- Morphological open + close
- Compute severity ratio (Low/Med/High)

Grad-CAM Explainability

- Build gradient model on out_relu layer
- Record forward pass in GradientTape
- Average-pool gradients → channel weights
- Overlay COLORMAP_JET heatmap on leaf

Environmental Analysis

- Temperature: 18–35°C optimal
- Humidity: 35–80% safe range
- Soil Moisture: 25–75% optimal
- Rule-based agronomic remarks

Dataset, Model & Image Enhancement

Dataset Preparation

- 13,460 leaf images across 18 classes
- 4 plant species: Neem, Tulsi, Turmeric, Aloe Vera
- Nested → flat directory via `flatten_dataset()`
- Split: 70% train / 20% val / 10% test (seed=42)
- Augmentation: $\pm 20^\circ$ rotate, flip, zoom 0.2, shift 0.1
- Pixel rescaling: /255 normalisation

MobileNetV2 Model

- Base:** MobileNetV2 (ImageNet)
- Input:** $224 \times 224 \times 3$
- Frozen:** Yes (2.26M params)
- Head:** GAP → Dense(128, ReLU)
- Dropout:** 0.3
- Output:** Dense(18, Softmax)
- Loss:** Categorical Cross-entropy
- Callbacks:** EarlyStopping, ReduceLR

CLAHE Enhancement

1. Resize to 224×224 px
2. BGR → RGB conversion
3. RGB → CIE LAB colour space
4. Extract L (lightness) channel
5. Apply CLAHE:
clipLimit=3.0, tile=8x8
6. Merge L + original A + B
7. LAB → RGB output

Segmentation, Explainability & Report Generation

Lesion Segmentation

- Background removal via grayscale thresholding ($T=25$)
- Morphological open + close to clean mask
- HSV masking: hue [10–35], saturation [30–255]
- Severity = lesion pixels / total pixels
- Low < 10% | Medium 10–30% | High > 30%

Grad-CAM Explainability

- Secondary Keras model outputs `out_relu` + predictions
- GradientTape records forward pass
- Gradient of class score w.r.t feature maps
- Global average-pool → channel importance weights
- COLORMAP_JET heatmap blended 60:40 with leaf image

Colour Statistics & Image QC

- Mean RGB & HSV channel statistics computed
- Greenness Index = $\text{mean}(G) - \text{avg}(\text{mean}(R), \text{mean}(B))$
- Leaf colour quality: Good / Moderate / Poor
- Image QC: resolution, brightness, Laplacian blur
- Flags images with confidence < 0.50

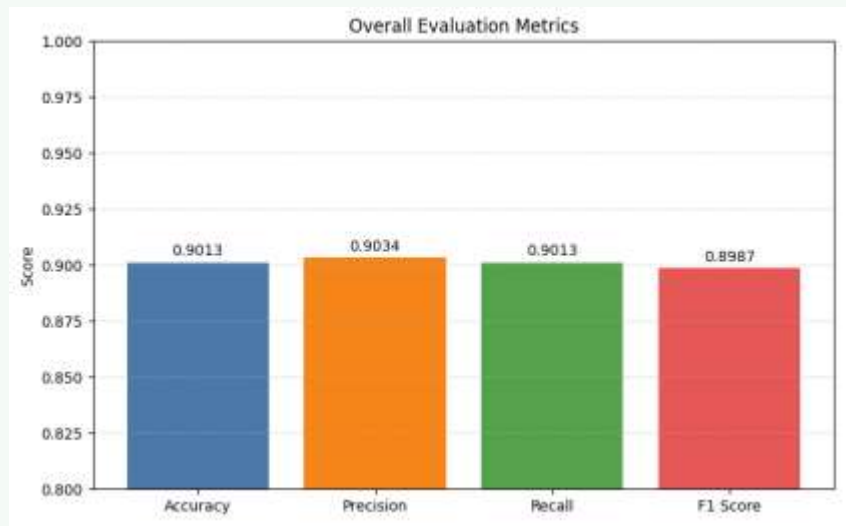
Diagnosis Report & Export

- 11-field pandas DataFrame per diagnosis run
- Plant-specific treatment via conditional logic tree
- Severity-graded advice: monitor → immediate action
- Export to Excel with `files.download()` in Colab
- 9-panel visual dashboard in web application

RESULTS & TESTING

5/7

Comprehensive Performance Analysis



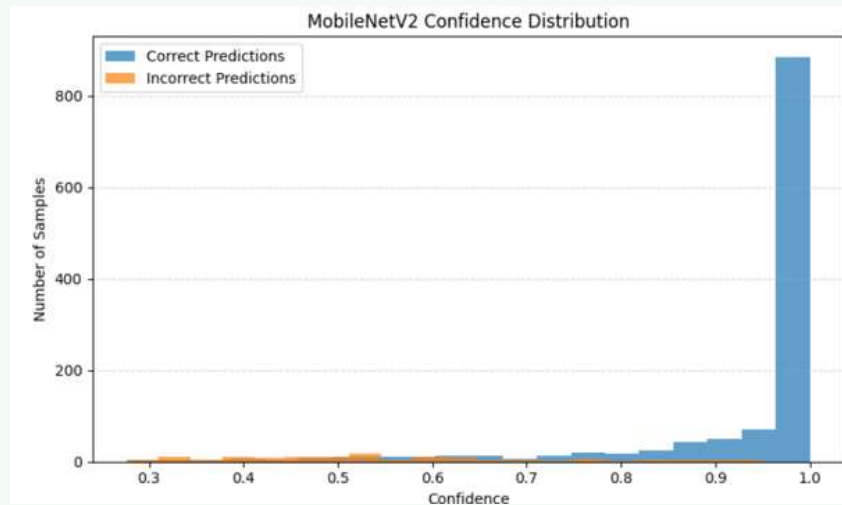
CONFIDENCE ANALYSIS

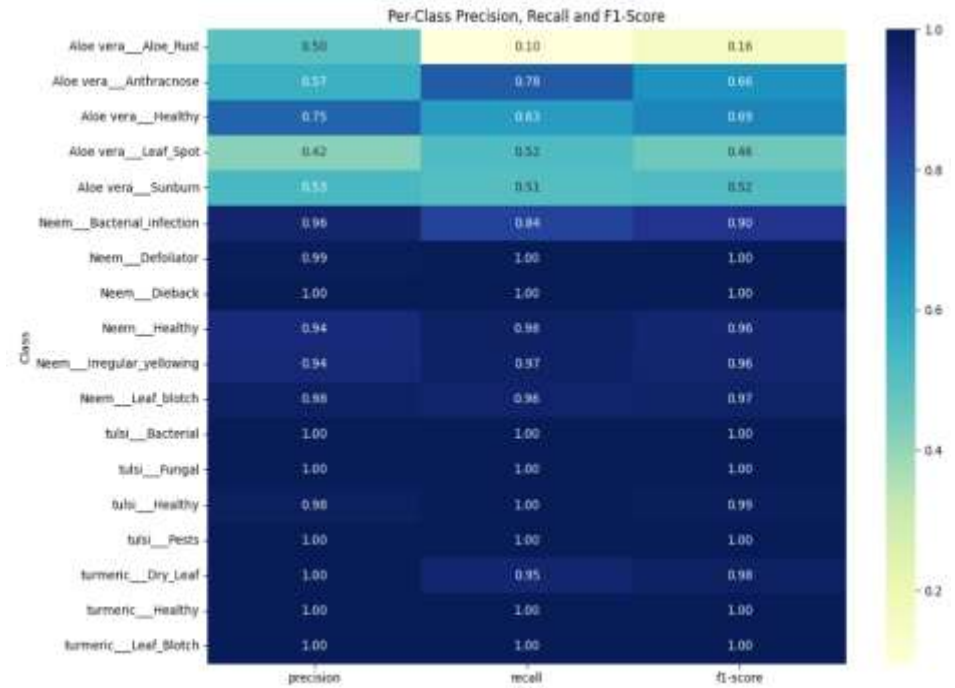
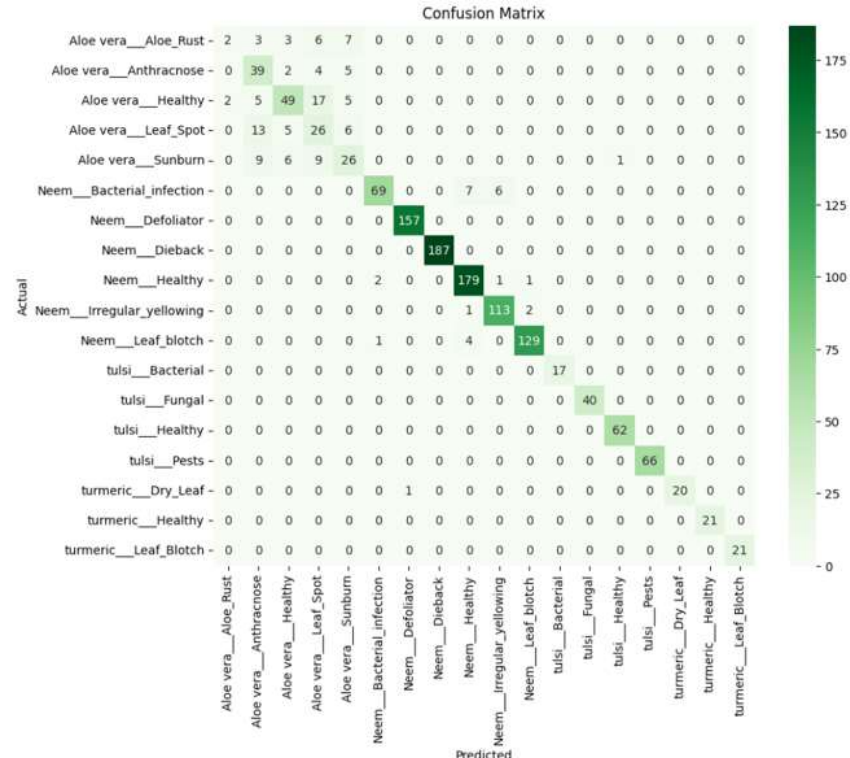
93.40%

Correct

55.20%

Incorrect





Original Image



Predicted Class: turmeric__Dry_Leaf
Confidence: 0.9033

Enhanced Image



Original



BG Removed



Enhanced



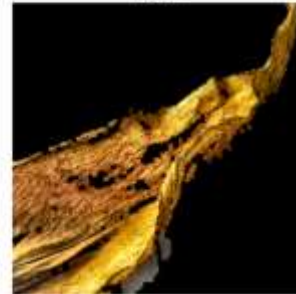
Lesion Mask



BG Mask



Lesion



Enhanced Image

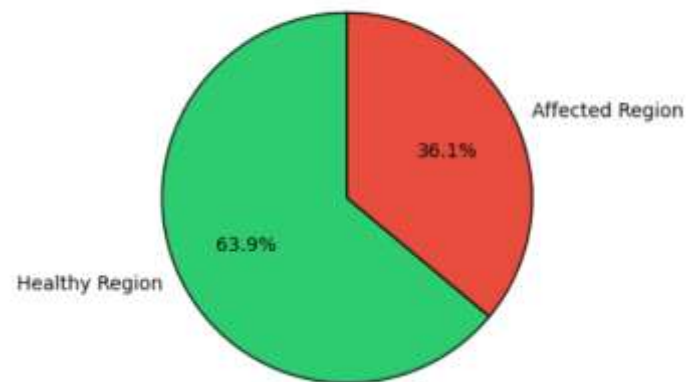


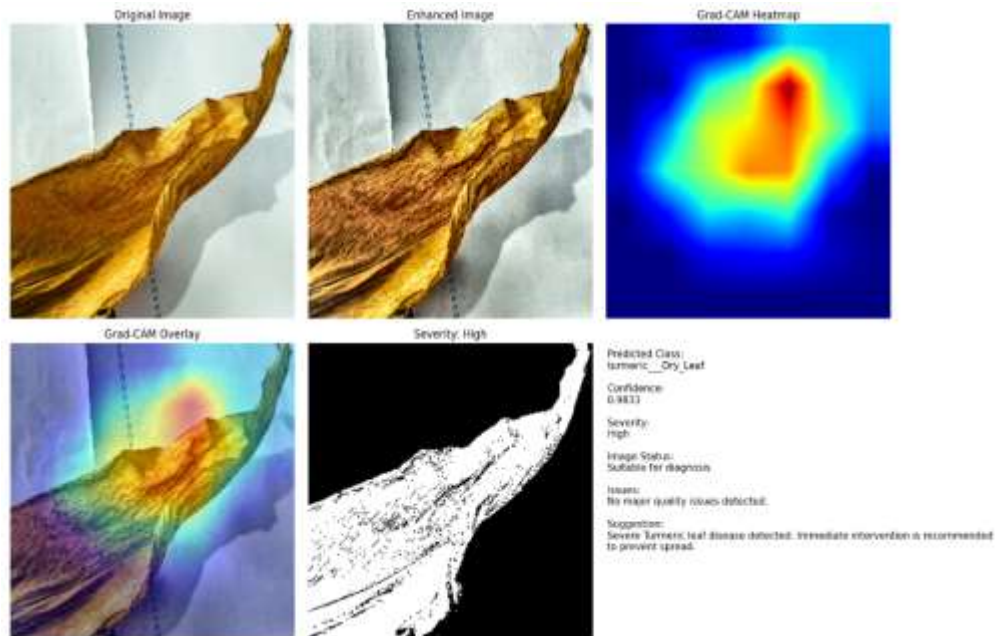
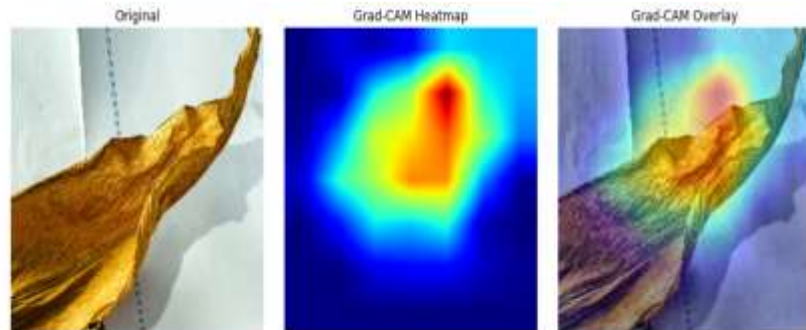
Severity: High
Severity Ratio: 0.3605

Estimated Severity: High
Ratio: 0.3605



Estimated Leaf Severity Distribution





AyurLeaf AI

Home

Analyze

Performance

Disease Library

AI POWERED MEDICINAL PLANT DASHBOARD

Smart Leaf Disease Detection for Herbal Plants

Upload a medicinal plant leaf image and get intelligent disease prediction, confidence score, health status, severity level, and plant care suggestions in a clean professional dashboard.

Analyze Leaf

View Performance

Upload Leaf Image

Select a medicinal plant leaf image to begin analysis.

Choose Image

Analyze Now



AyurLeaf AI

Home

Analyze

Performance

Disease Library

LIVE DASHBOARD

Analyze Plant Leaf

Upload a clear leaf image and let the model generate a disease prediction, confidence score, health status, and care suggestion.

Upload Leaf Image

Select a medicinal plant leaf image to begin analysis.

Choose Image

Analyze Now

PREDICTED DISEASE

PREDICTED DISEASE

turmeric → Dry Leaf

96.53%

Affected

High Severity

Health Status

Affected

Severity

High

Color Quality

Possible discoloration detected

Suggestion

Maintain proper soil moisture and monitor environmental stress conditions.

Prediction Probability



MODEL INSIGHTS

Performance Dashboard

Accuracy

90.13%

Precision

90.34%

Recall

90.13%

F1-Score

89.87%

Confidence Comparison



Training Accuracy Curve



Training Loss Curve



Health Overview



AyurLeaf AI

Home Analysis Performance Disease Library

KNOWLEDGE BASE

Plant Information & Disease Library

Tomato

Common Diseases: Fusarium Wilt, Late Blight, Septoria, Bacterial Spot, Yellow Leaf Curl.

Symptoms: Wilting, leaf damage, brown drying, brown patches.

Care Tip: Monitor moisture, watering, proper infection prevention.

Preventive Advice: Regularly inspect leaves and remove affected regions early.

Leaf

Common Diseases: Rust, Powdery Mildew, Spider Mites.

Symptoms: White patches, dark spots, yellowing, curling leaves.

Care Tip: Keep in sunlight, avoid overwatering, use fungicide.

Preventive Advice: Remove excess moisture and ensure proper airflow.

Aloe Vera

Common Diseases: Mealybugs, Aphids, Scale, Root Rot.

Symptoms: Brown spots, yellowing, distorted edges, soft spots.

Care Tip: Avoid overwatering, remove damaged leaves.

Preventive Advice: Keep leaves dry and protect from insect attacks.

Banana

Common Diseases: Dry Leaf Spot, Leaf Blight.

Symptoms: Dry leaf tips, brown spots, reduced yield quality.

Care Tip: Keep soil moist, support with stakes, monitor moisture.

Preventive Advice: Use healthy planting material and detect symptoms early.

CONCLUSION & FUTURE SCOPE

CONCLUSION

90.13% Accuracy

On 18-class Ayurvedic plant disease dataset — exceeds 85% target

Lightweight Model

9.25 MB MobileNetV2 — deployable on mobile & edge devices

CLAHE Preprocessing

Improved lesion visibility without distorting disease colour signatures

Pixel-Level Severity

HSV segmentation classifies Low / Medium / High disease extent

Grad-CAM Explainability

Spatial heatmaps confirm model attends to disease lesions

Complete Web App

AyurLeaf AI — Home, Analyze, Performance, Disease Library pages

Exportable Reports

11-field Excel diagnosis with plant-specific treatment guidance

FUTURE SCOPE

1

Dataset Expansion

Add Ashwagandha, Brahmi, Giloy; collect more Turmeric disease images

2

Fine-Tune Base Layers

Gradual unfreezing of MobileNetV2 upper layers for intra-species diseases

3

U-Net Lesion Segmentation

Replace rule-based HSV with semantic segmentation for precision

4

IoT Sensor Integration

Auto-populate temperature, humidity & soil moisture from field sensors

5

TFLite Mobile App

On-device inference for rural areas with limited internet connectivity

6

Temporal Disease Tracking

Monitor disease progression over time; validate treatment effectiveness

7

Standards Alignment

Align severity thresholds to BIS / ICAR / FAO agronomic standards

REFERENCES

Key Literature & Standards

[1] **A. Mohanadevi et al.**

Enhanced Medicinal Plants Identification Using CNN & MobileNetV2

ICCCES 2026

[2] **V. Tanwar et al.**

Enhanced Classification of Turmeric Leaf Disease Severity Using Hybrid Deep Learning

ICEMPS 2024

[3] **A. Iqbal**

PlantHealthNet: Transformer-Enhanced Hybrid Models for Disease Diagnosis

IEEE Access 2025

[4] **J. Lande et al.**

Transfer Learning-Based Plant Leaf Disease Classification Using MobileNetV2

ICAAIC 2025

[5] **A. R. Revathi et al.**

Efficient Transfer Learning for Identification & Severity Prediction of Plant Diseases

ICCSPP 2025

[6] **R. Kumar et al.**

Plant Leaf Diseases Severity Estimation Using Fine-Tuned CNN Models

ISCON 2023

[7] **P. Kambli et al.**

Identification of Ayurvedic Plants Using Deep Learning

COSMIC 2024

[8] **K. Saini & P. Agarwal**

Aloe Vera Disease Prediction Using Machine Learning Techniques

ICOSEC 2024

[9] **S. S. Patil et al.**

Tulsi Leaf Disease Detection Using CNN

IEEE IATMSI 2022

[10] **J. B. Guptha & E. Elakiya**

Detection of Neem Leaf Disease Using Deep Learning Models

ICDSBS 2025

THANK YOU